Inferring Coercions for Monadic Programming

This paper shows how to use subtyping to infer uses of the basic monad operations.

CCS Concepts: • **Software and its engineering → Functional languages**; *Recursion.*

Additional Key Words and Phrases: strong functional programming, type-based termination, Mendler-style recursion

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1 INTRODUCTION

The basic declarative subtyping rules for applicatives are shown in Figure 1. Consider this example subtyping:

$$\mathsf{N} \to \mathsf{N} \quad <: \quad \mathsf{L}\,\mathsf{N} \to \mathsf{L}\,(\mathsf{L}\,\mathsf{N})$$

This is valid for applicatives, because we may first use `fmap` (which follows from `pure` and `seq`) to add L to the domain and codomain parts of the type, and then we may use function subtyping to expose the codomain part, and add an extra L with `pure`. So the coercion of $f : \mathsf{N} \to \mathsf{N}$ would be

$$\mathit{pure} \cdot (\mathit{fmap}\ f)$$

Note that with our current set of coercions, we cannot apply the function subtyping rule at the beginning, because we do not have (contravariantly) $\mathsf{L}\,\mathsf{N} <: \mathsf{N}$. So we are forced to use composition (transitivity) here.

The challenge of admissibility of transitivity may be thought of as formulating a set of coercion combinators excluding composition. We need some form of function subtyping rule that takes occurrences of L in the domain part of the upper bound type (T_2 in $T_1 <: T_2$) into account. To do this, we adopt a new function subtyping rule that keeps track of how many applications of the functor would need to be inserted.

2 A DECISION PROCEDURE

We work with types T and T' in the syntax shown in Figure 2. We use N as a base type (it could be the natural numbers), and L as an example covariant functor (like the list datatype). The figure also gives the syntax for *modes* θ, which play an important role in our decision procedure for subtyping. The mode τ means a top-level position in the types, while a means any position. Figure 3 presents that procedure in the form of a proof system for deriving judgments of the form $(k)\ T <:^{\theta} T'$, where $k \in \mathbb{N}$. The semantics of this judgment is given by the following definition.

Definition 1.

- $[\![(k)\ T <:^{a} T']\!] = \mathsf{L}^k\ T <: T'$

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$$\frac{}{\mathsf{L}\,(T_1 \to T_2) <: \mathsf{L}\,T_1 \to \mathsf{L}T_2}\ \mathit{seq} \qquad \frac{}{T <: \mathsf{L}\,T}\ \mathit{pure}$$

$$\frac{T <: T'}{\mathsf{L}\,T <: \mathsf{L}\,T'}\ \mathit{cov} \qquad \frac{T_1' <: T_1 \quad T_2 <: T_2'}{T_1 \to T_2 <: T_1' \to T_2'}\ \mathit{fun}$$

$$\frac{T_1 <: T_2 \quad T_2 <: T_3}{T_1 <: T_3}\ \mathit{trans} \qquad \frac{}{\mathsf{N} <: \mathsf{N}}\ \mathit{refl}$$

Fig. 1. Declarative subtyping for applicatives

$$\begin{array}{llll} \textit{Types} & S, T & ::= & \mathsf{N} \mid \mathsf{L}\,T \mid T \to T' \\ \textit{Modes} & \theta & ::= & a \mid \tau \end{array}$$

Fig. 2. Syntax of types

$$\frac{}{(0)\ \mathsf{N} <:^a \mathsf{N}}\ \mathit{refl} \qquad \frac{(j)\ T_1' <:^\tau T_1 \quad (k)\ T_2 <:^a T_2' \quad i \le j \le k}{(i)\ T_1 \to T_2 <:^a T_1' \to T_2'}\ \mathit{fun}$$

$$\frac{(0)\ T <:^a T'}{(0)\ T <:^a \mathsf{L}\,T'}\ \mathit{pure} \qquad \frac{(0)\ T <:^a T'}{(0)\ T <:^\tau T'}\ \mathit{reset}$$

$$\frac{(k+1)\ T <:^a T'}{(k)\ \mathsf{L}\,T <:^a T'}\ \mathit{seq\text{-}cod} \qquad \frac{(k)\ T <:^a T'}{(k+1)\ T <:^a \mathsf{L}\,T'}\ \mathit{pull}$$

$$\frac{(k)\ T <:^\tau T'}{(k+1)\ \mathsf{L}\,T <:^\tau T'}\ \mathit{seq\text{-}dom}$$

Fig. 3. Proposed rules for applicatives

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{(0)\ \mathsf{N} <:^a \mathsf{N}}}{(0)\ \mathsf{N} <:^\tau \mathsf{N}}}{(1)\ \mathsf{L}\,\mathsf{N} <:^\tau \mathsf{N}}\ \mathit{seq\text{-}dom} \quad \dfrac{\dfrac{}{(0)\ \mathsf{N} <:^a \mathsf{N}}}{(1)\ \mathsf{N} <:^a \mathsf{L}\,\mathsf{N}}\ \mathit{pure}}{(1)\ \mathsf{N} \to \mathsf{N} <:^a \mathsf{L}\,\mathsf{N} \to \mathsf{L}\,\mathsf{N}}}{(0)\ \mathsf{L}\,(\mathsf{N} \to \mathsf{N}) <:^a \mathsf{L}\,\mathsf{N} \to \mathsf{L}\,\mathsf{N}}\ \mathit{seq\text{-}cod} \quad \dfrac{\dfrac{\dfrac{}{(0)\ \mathsf{N} <:^a \mathsf{N}}}{(1)\ \mathsf{N} <:^a \mathsf{L}\,\mathsf{N}}\ \mathit{pure}}{(1)\ \mathsf{L}\,\mathsf{N} <:^a \mathsf{L}\,(\mathsf{L}\,\mathsf{N})}\ \mathit{cov}}{(0)\ \mathsf{L}\,(\mathsf{N} \to \mathsf{N}) <:^\tau \mathsf{L}\,\mathsf{N} \to \mathsf{L}\,\mathsf{N}}$$

$$(0)\ (\mathsf{L}\,\mathsf{N} \to \mathsf{L}\,\mathsf{N}) \to \mathsf{L}\,\mathsf{N} <:^a \mathsf{L}\,(\mathsf{N} \to \mathsf{N}) \to \mathsf{L}\,(\mathsf{L}\,\mathsf{N})$$

Fig. 4. Example derivation

- $[\![(k)\ T <:^\tau T']\!] = \exists\, T''.\ T = \mathsf{L}^k\,T'' \ \wedge\ T'' <: T'$

We prove below that the rules are sound with respect to this semantics. An example derivation with the rules of Figure 3 is shown in Figure 4.

3 SOUNDNESS

Theorem 2. *If* $(k)\ T <:^{\theta} T'$ *is derivable, then* $[\![(k)\ T <:^{\theta} T']\!]$.

Proof. The proof is by induction on the structure of the derivation.

Case:

□

REFERENCES

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